SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1023	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – **Agile Sprint Planning & User Story Workshop**
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
Student Name:		**Reg. No.:**	
Date:		**Duration:**	60 minutes

Code of Conduct

I M. Varun Sai / 192425247 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: **M. Varun Sai**

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

1. **Identify the Project Vision and Stakeholders**
   - Define the project's vision, objectives, and identify all key stakeholders involved.
2. **Create Epics and User Stories**
   - Break down the requirements into Epics and formulate well-structured User Stories using the format: *As a <user>, I want <feature>, so that <benefit>.*
3. **Define Acceptance Criteria**
   - Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.
4. **Prioritize the Product Backlog**
   - Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.
5. **Plan the Sprint Backlog**
   - Select high-priority user stories for the first sprint and divide them into actionable development tasks.
6. **Estimate Effort Using Story Points**
   - Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.
7. **Present the Sprint Goal and Expected Deliverables**
   - Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

1. Project Vision and Stakeholders

1.1 Project Vision

The vision of this project is to develop a centralized web-based platform that connects consumers, collection centres, logistics providers, and recyclers to manage e-waste collection, transportation, recycling, and sustainability reporting efficiently.

1.2 Project Objectives

- Simplify e-waste collection requests and pickup scheduling.
- Track e-waste from collection to recycling.
- Maintain accurate classification and recycling records.
- Provide sustainability dashboards and reports.
- Improve security, scalability, and reliability using modern software engineering practices.
- Reduce manual coordination and encourage responsible disposal.

1.3 Key Stakeholders

Stakeholder	Responsibilities
Consumers	Submit collection requests and track their status.
Collection Centres	Receive, classify, and store e-waste.
Logistics Providers	Schedule pickups and transport e-waste.
Recyclers	Process e-waste and record recovered materials.
Administrators	Manage users, requests, and system activities.
Government Authorities	Monitor regulatory compliance and environmental reporting.
Development Team	Design, develop, test, and maintain the platform.
Product Owner	Define requirements and prioritize the backlog.
Scrum Master	Facilitate Scrum activities and remove team impediments.

2. Create Epics and User Stories

An Epic is a large feature that can be divided into smaller user stories.

Epic 1: User Management

- US1: As a consumer, I want to register and log in so that I can access the platform securely.
- US2: As an administrator, I want to manage user accounts so that only authorized users can access the system.

Epic 2: E-Waste Collection

- US3: As a consumer, I want to submit an e-waste collection request so that my unwanted electronic devices can be collected.
- US4: As a consumer, I want to schedule a pickup so that e-waste can be collected at a convenient time.

- US5: As a logistics provider, I want to view assigned pickups so that I can organize collection activities.

Epic 3: Tracking and Classification

- US6: As a consumer, I want to track my collection request so that I know its current status.
- US7: As a collection-centre operator, I want to classify e-waste by type and weight so that materials can be processed correctly.

Epic 4: Recycling Management

- US8: As a recycler, I want to record recycling activities so that material recovery can be monitored.
- US9: As an administrator, I want to view recycling records so that I can verify recycling activities.

Epic 5: Sustainability Analytics

- US10: As an administrator, I want to view sustainability reports so that I can measure collection and recycling performance.

Epic 6: System Monitoring

- US11: As a system operator, I want to monitor application health and errors so that I can maintain reliable services.
- US12: As a developer, I want automated testing and deployment so that software updates can be released consistently.

3. Define Acceptance Criteria

Acceptance criteria specify the conditions that must be satisfied for a user story to be accepted.

US1: User Registration and Login

- Given a new user provides valid registration details, when they submit the form, then an account should be created.
- Given a registered user enters valid credentials, when they log in, then they should access the appropriate dashboard.
- Given invalid credentials are entered, when login is attempted, then an error message should be displayed.

US3: E-Waste Collection Request

- Given a logged-in consumer, when they enter the device type, quantity, pickup address, and contact details, then the system should save the request.
- Given required information is missing, when the request is submitted, then validation errors should be displayed.
- Given a request is saved successfully, then the system should generate a unique request ID.

US4: Pickup Scheduling

- Given an existing collection request, when the consumer selects an available pickup slot,

then the system should save the selected date and time.

- Given a pickup slot is unavailable, when the consumer attempts to select it, then the system should prevent the booking and request another slot.

US6: Request Tracking

- Given a valid collection request ID, when the consumer checks its status, then the system should display the latest recorded status.
- Given the logistics provider updates the status, when the update is saved, then the consumer should see the updated information.

US7: E-Waste Classification

- Given a collection-centre operator is authenticated, when they enter the e-waste category and weight, then the system should save the classification record.
- Given invalid or missing weight information, when the record is submitted, then the system should display a validation message.

US8: Recycling Records

- Given a recycler receives e-waste, when the recycler enters the processed quantity and recovered materials, then the system should save the recycling record.
- Given a record is saved, then authorized users should be able to view it.

US10: Sustainability Reports

- Given collection and recycling records exist, when an administrator opens the sustainability dashboard, then the system should display total collected weight, recycled weight, and recovery statistics.
- Given no records exist for the selected period, then the dashboard should display zero values or an appropriate no-data message.

4. Prioritize the Product Backlog

The product backlog is prioritized using the MoSCoW technique: Must Have, Should Have, Could Have, and Won't Have for the current release.

Priority	User Story	Business Value
Must Have	US1: User registration and login	Enables secure access.
Must Have	US3: Collection request	Provides the main platform function.
Must Have	US4: Pickup scheduling	Enables collection coordination.
Must Have	US6: Request tracking	Provides status visibility.
Should Have	US7: E-waste classification	Supports correct material handling.
Should Have	US5: Assigned pickup management	Improves logistics coordination.
Should	US8: Recycling records	Tracks recycling activities.

Have		
Should Have	US2: User administration	Supports account management.
Could Have	US10: Sustainability dashboard	Supports environmental analysis.
Could Have	US9: Recycling record review	Improves administrative oversight.
Should Have	US11: System monitoring	Improves operational reliability.
Should Have	US12: Automated testing and deployment	Improves release quality.

Prioritization justification: User authentication, collection requests, and pickup scheduling are prioritized because they establish the platform's core workflow. Tracking follows to provide visibility. Classification, recycling, and analytics can be expanded after the basic collection process is working.

5. Plan the Sprint Backlog

Sprint duration: 2 weeks

Sprint number: Sprint 1

Selected stories: US1, US3, and US4, with basic project setup and testing tasks.

Sprint Backlog and Development Tasks

User Story	Development Tasks	Responsible Role
US1: Registration and login	Design registration/login pages; create user database; implement authentication; validate inputs; test login scenarios.	Frontend and backend developers
US3: Collection request	Design request form; create request database table; implement request submission API; generate request ID; test validation.	Frontend and backend developers
US4: Pickup scheduling	Design pickup-slot interface; create scheduling database table; implement slot validation; save bookings; test slot conflicts.	Frontend and backend developers
Project setup	Configure repository; establish development environment; create initial database schema; configure basic automated tests.	Development team
Testing and integration	Test registration, request creation, scheduling, and integration between modules.	Testing team

Sprint Workflow

1. Conduct sprint planning and confirm the selected stories.
2. Design the database, user interface, and API endpoints.

3. Develop the selected features.
4. Integrate and test the modules.
5. Demonstrate the working features during the sprint review.
6. Conduct a retrospective and identify improvements for Sprint 2.

6. Estimate Effort Using Story Points

Story points estimate the relative effort, complexity, and uncertainty of a user story. The team uses the Fibonacci sequence: 1, 2, 3, 5, 8, 13.

6.1 Story Point Estimates

User Story	Story Points	Justification
US1: Registration and login	5	Requires authentication, database integration, and validation.
US3: Collection request	5	Requires form validation, API development, and request ID generation.
US4: Pickup scheduling	8	Requires slot management, availability validation, and conflict handling.
Total	18	Estimated Sprint 1 workload

6.2 Estimation Technique

The team uses Planning Poker, where each team member independently estimates the complexity of each story using Fibonacci numbers. The team discusses differences and agrees on a final estimate.

The scheduling story receives 8 points because it involves more complex validation and coordination than a basic registration or request form.

The total of 18 points is a proposed estimate, not a guaranteed delivery capacity. The team should confirm it against its previous velocity or available working time.

7. Sprint Goal and Expected Deliverables

7.1 Sprint Goal

To develop a working initial version of the platform that allows consumers to register securely, submit e-waste collection requests, and schedule pickup slots.

7.2 Expected Deliverables

At the end of Sprint 1, the team expects to deliver:

1. A working registration and login interface.
2. Secure user authentication and basic validation.
3. An e-waste collection request form.
4. A database for storing collection requests.
5. Automatic generation of unique request IDs.

6. A pickup scheduling interface.
7. Basic pickup-slot availability and conflict validation.
8. Integrated modules tested against the defined acceptance criteria.
9. Source code maintained in a version-control repository.
10. A sprint review demonstration and testing report.

7.3 Definition of Done

A user story is considered complete when:

- All acceptance criteria are satisfied.
- Code is reviewed and integrated.
- Required tests pass.
- Input validation and error handling work correctly.
- No critical defects remain open.
- The feature is demonstrated to the Product Owner.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification (30%)	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories (25%)	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria (20%)	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning (15%)	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation (10%)	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification **(30%)**	/5
Epics & User Stories **(25%)**	/5
Acceptance Criteria **(20%)**	/5
Product Backlog Prioritization & Sprint Planning **(15%)**	/5
Story Point Estimation, Sprint Goal & Presentation **(10%)**	/5
TOTAL	/25